

API Roadmap.

PHASE-I

Start from basics (Obvious).

- * API
- * Working of API
- * API Types
- * Request/Response Cycle
- * JSON
- * Statelessness

PHASE-II

Master HTTP (Must) **

- * HTTP
- * HTTP Versions
- * HTTP Methods
- * HTTP Status Codes
- * HTTP Headers
- * Cookies
- * CORS
- * HTTP Caching

Use Postman or Curl & Public APIs.

PHASE-III

API Styles

- REST **
- GraphQL **
- WebSockets **
- JSON API
- SOAP API
- gRPC API

PHASE - IV

Authentication & Security

API without security = broken system.

- Basic Auth
 - Token based Auth (JWT)
 - OAuth 2.0
 - Role based access control
 - Attribute based Access Control
- * ABAC
 - * DAC
 - * PBAI
 - * RBAC
 - * MAC
 - * ReBAC

PHASE-V

API Features

- Rest Principle
 - URL Design **
 - Versioning strategies
 - Handling CRUD **
 - Basics of DNS
 - Content Negotiation
 - Query & path parameters
 - Pagination **
 - Rate limiting. **
 - HATEOAS
 - Throttling
- Node.js → Express
Python → FastAPI/Flask

PHASE-VI API Performance.

- performance Metrics
- Load Balancing
- Profiling and Monitoring
- Error Handling / Retries.
- performance testing
- Basic Attack factors
- Routing

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PHASE-VII API Integration Patterns

- * Synchronous Vs Asynchronous APIs.
- * Event driven Architecture.
- * Webhook Vs polling.
- * Batch processing
- * Messaging Queues
- * Microservice API.

Kafka
RabbitMQ

Additional

- * Best practices to use API
- * Mocking APIs
- * API Contracts
- * PCI DSS.
- * CCPA

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* PII

* HIPAA

